$H^i = \mathrm{Ker}(d_i)/\mathrm{Im}(d_{i-1})$. The complex is said to be **exact** in the ith term if $H^i = 0$ and is said to be **an exact sequence** if it is exact in all terms.

There is an obvious notion of a **morphism of complexes**. Namely, a morphism $f : C \to D$ is a collection of morphisms $f_i : C_i \to D_i$ such that $d_i \circ f_i = f_{i+1} \circ d_i$. Clearly, such morphisms can be composed, which makes the class of all complexes over $\mathcal{C}$ into a category, denoted $\mathrm{Compl}(\mathcal{C})$.

In particular, one can consider complexes of abelian groups, vector spaces, modules over a ring, etc. In this case, elements of $\mathrm{Ker}(d_i)$ are called i-**cocycles**, the elements of $\mathrm{Im}(d_{i-1})$ are called i-**coboundaries**, and the elements of $H^i(C)$ are called ith **cohomology classes**. This terminology is adopted from topology. For this reason, exact sequences are also called **acyclic complexes** (as they are complexes which have no nontrivial cocycles, i.e., cocycles that are not coboundaries).

Often one considers complexes that are bounded in one or both directions; i.e., the objects C_i are zero for $i \gg 0$, $i \ll 0$, or both. In this case one writes one zero on each side where the complex is bounded. For example, a complex bounded on both sides with $n+1$ terms will look like

$$0 \to C_0 \to C_1 \to \cdots \to C_n \to 0.$$

Definition 7.8.2. A **short exact sequence** is an exact sequence of the form

$$0 \to X \to Y \to Z \to 0.$$

Clearly, $0 \to X \to Y \to Z \to 0$ is a short exact sequence if and only if $X \to Y$ is injective, $Y \to Z$ is surjective, and the induced map $Y/X \to Z$ is an isomorphism. In other words, short exact sequences correspond to extensions of Z by X.

Example 7.8.3. The sequence $0 \to X \to X \oplus Z \to Z \to 0$ with the obvious morphisms is a short exact sequence. Such a sequence is called **split**. It corresponds to the trivial extension of Z by X.